Appendix: Accessibility, Cognitive Ergonomics, and Participatory Research Infrastructure

Automated meta-analysis tools operate at the intersection of computational infrastructure and human sensemaking. The scalability gains demonstrated by the present pipeline are meaningful only if the resulting knowledge artefacts remain cognitively accessible, ethically transparent, and open to diverse forms of participation. This appendix situates our work within the broader landscape of research accessibility, cognitive ergonomics, decentralized science (DeSci), and participatory infrastructure design.

Cognitive Ergonomics of Knowledge Graphs

The knowledge graph outputs of this pipeline—hypothesis dashboards, citation networks, temporal evidence trajectories—impose nontrivial cognitive demands on users who must interpret multidimensional evidence landscapes. Cognitive Load Theory sweller2011cognitive establishes that information system designs which exceed working-memory capacity produce